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Aim : Aim: Performance Evaluation of MANET Protocols

Measure throughput, packet delivery ratio, and end-to-end delay for different MANET routing protocols.

```
# =====
# MANET Simulation using AODV
# NS-2.35
# =====

# Create simulator
set ns [new Simulator]

# -----
# Trace files
# -----

set tracefd [open manet.tr w]
$ns trace-all $tracefd

set namfd [open manet.nam w]
$ns namtrace-all-wireless $namfd 800 800

# -----
# Topography
# -----

set topo [new Topography]
$topo load_flatgrid 800 800

# Create GOD for 5 nodes
create-god 5

# -----
# Wireless channel
# -----

set chan_1_ [new Channel/WirelessChannel]

# -----
# Node configuration
# -----

$ns node-config \
    -adhocRouting AODV \
```

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```
-llType LL \  
-macType Mac/802_11 \  
-ifqType Queue/DropTail/PriQueue \  
-ifqLen 50 \  
-antType Antenna/OmniAntenna \  
-propType Propagation/TwoRayGround \  
-phyType Phy/WirelessPhy \  
-channel $chan_1_ \  
-topoInstance $topo \  
-agentTrace ON \  
-routerTrace ON \  
-macTrace ON \  
-movementTrace ON
```

```
# -----  
# Create 5 nodes  
# -----
```

```
for {set i 0} {$i < 5} {incr i} {  
    set node_($i)[ns node]  
}
```

```
# -----  
# Initial positions  
# -----
```

```
$node_(0) set X_ 100  
$node_(0) set Y_ 100  
$node_(0) set Z_ 0
```

```
$node_(1) set X_ 250  
$node_(1) set Y_ 200  
$node_(1) set Z_ 0
```

```
$node_(2) set X_ 400  
$node_(2) set Y_ 300  
$node_(2) set Z_ 0
```

```
$node_(3) set X_ 550  
$node_(3) set Y_ 400  
$node_(3) set Z_ 0
```

```
$node_(4) set X_ 700  
$node_(4) set Y_ 500
```

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```
$node_(4) set Z_ 0
```

```
# -----
```

```
# Set initial node positions in NAM
```

```
# -----
```

```
for {set i 0} {$i < 5} {incr i} {  
    $ns initial_node_pos node_$i 20  
}
```

```
# -----
```

```
# UDP Agent
```

```
# Source = Node 0
```

```
# -----
```

```
set udp0 [new Agent/UDP]
```

```
$ns attach-agent $node_(0) $udp0
```

```
# -----
```

```
# Null Agent
```

```
# Destination = Node 4
```

```
# -----
```

```
set null0 [new Agent/Null]
```

```
$ns attach-agent $node_(4) $null0
```

```
# -----
```

```
# Connect UDP to Null
```

```
# -----
```

```
$ns connect $udp0 $null0
```

```
# -----
```

```
# CBR Traffic
```

```
# -----
```

```
set cbr0 [new Application/Traffic/CBR]
```

```
$cbr0 set packetSize_ 512
```

```
$cbr0 set interval_ 0.1
```

```
$cbr0 attach-agent $udp0
```

```
# -----
```

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```
# Mobility
# -----

nsat1.0"node_(1) setdest 300 300 10"
nsat2.0"node_(2) setdest 450 350 10"
nsat3.0"node_(3) setdest 600 450 10"

# -----
# Start and stop CBR traffic
# -----

nsat1.0"cbr0 start"
nsat8.0"cbr0 stop"

# -----
# Finish procedure
# -----

proc finish {} {
    global ns tracefd namfd

    $ns flush-trace

    close $tracefd
    close $namfd

    puts "Simulation completed."
    puts "Trace file: manet.tr"
    puts "NAM file: manet.nam"

    exec nam manet.nam &
    exit 0
}

# -----
# End simulation
# -----

$ns at 10.0 "finish"

puts "Simulation Started..."
$ns run
```

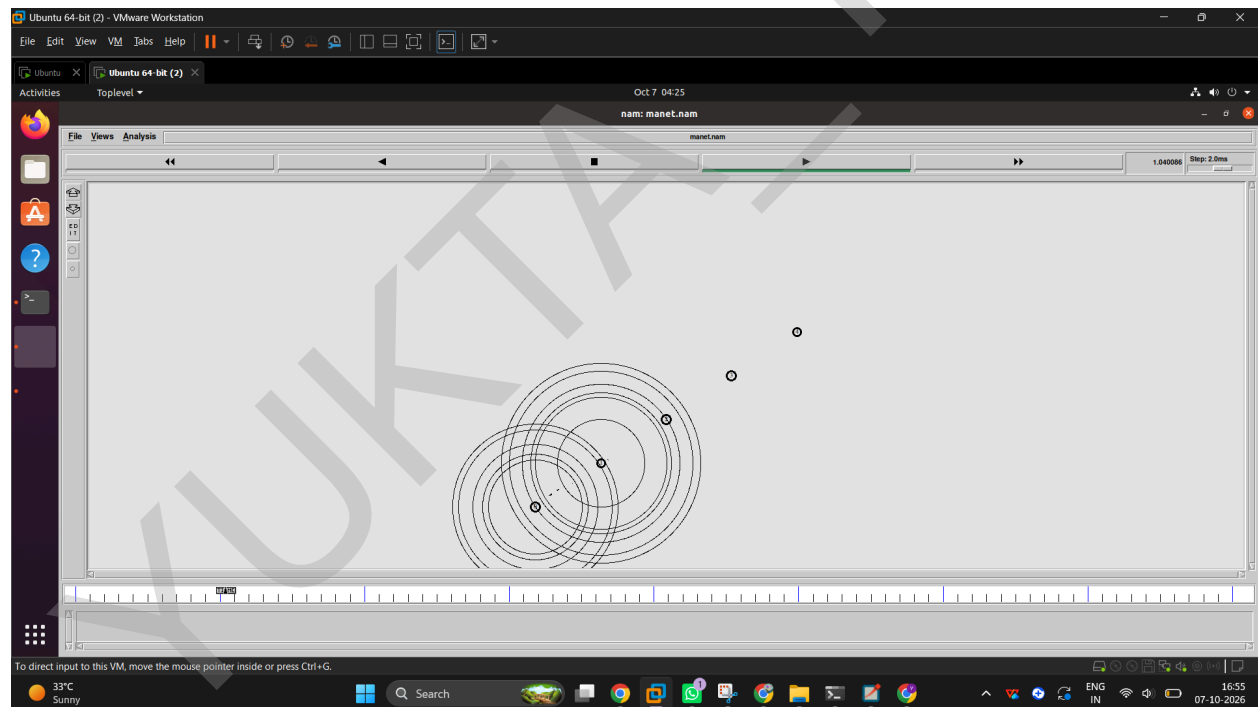
```
Ubuntu 64-bit (VMware Workstation)
File Edit View VM Tabs Help
Activities Terminal Oct 7 04:26 student@ubuntu:~/Downloads

student@ubuntu:~/Downloads$ sudo find -type f -name "onnetpp-6.4.0-linux-x86_64.tgz" > /dev/null
student@ubuntu:~/Downloads$ cd ~/Downloads
student@ubuntu:~/Downloads$ wget https://github.com/onnetpp/onnetpp/releases/download/onnetpp-6.4.0/onnetpp-6.4.0-linux-x86_64.tgz
--2026-10-07 03:42:53-- https://github.com/onnetpp/onnetpp/releases/download/onnetpp-6.4.0/onnetpp-6.4.0-linux-x86_64.tgz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)[20.207.73.82]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/160219599/ead7995c-f7f8-4fd0-ab22-361aa54bb4c4?sp=r&sv=2018-11-09&s=b&sr=h&tpsase=2026-10-07T11K3A21N3KA5Z7Rscd-at
tachment%3Bfflenane%3Donnetpp-6.4.0-linux-x86_64.tgz&rct=application%2Foctet-stream&skold=9cc2d410-5711-43a1-aedd-ab1947aa7ab8akclde-398a6d54-997b-47e9-b12b-9515b89db4dedaskst-2026-10-07T10K3A21N3KA09Z&sk
taehv1dxlcmnbnlbqnu291tlzV51jola2VSM5IsmV4CCTMTCSMTHMZM3MSwslbmJniJoXnzKmZYSNZ1CLCWXYKoiJotcnmZWZFZFWZCZVCbZJVHJdLvb1StbhG1MLNVCMUudLUZG93csyUZQXlfQ_GVL163tnPKpiYfUVLGrtIG2MHNBCNSessP
IFYfVMResponse-content-disposition=attachment%3Bfflenane%3Donnetpp-6.4.0-linux-x86_64.tgz&rct=response-content-type=application%2Foctet-stream [following]
--2026-10-07 03:42:53-- https://release-assets.githubusercontent.com/github-production-release-asset/160219599/ead7995c-f7f8-4fd0-ab22-361aa54bb4c4?sp=r&sv=2018-11-09&s=b&sr=h&tpsase=2026-10-07T11K3A2
1N3KA5Z7Rscd-at%3Bfflenane%3Donnetpp-6.4.0-linux-x86_64.tgz&rct=application%2Foctet-stream&skold=9cc2d410-5711-43a1-aedd-ab1947aa7ab8akclde-398a6d54-997b-47e9-b12b-9515b89db4dedaskst-2026-10-07T1
0K3A21N3KA09Z&sktaehv1dxlcmnbnlbqnu291tlzV51jola2VSM5IsmV4CCTMTCSMTHMZM3MSwslbmJniJoXnzKmZYSNZ1CLCWXYKoiJotcnmZWZFZFWZCZVCbZJVHJdLvb1StbhG1MLNVCMUudLUZG93csyUZQXlfQ_GVL163tnPKpiYfUVLGrtIG2MHNBCNSessP
IFYfVMResponse-content-disposition=attachment%3Bfflenane%3Donnetpp-6.4.0-linux-x86_64.tgz&rct=response-content-type=application%2Foctet-stream
Receiving release-assets.githubusercontent.com (release-assets.githubusercontent.com)[185.199.110.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 422878463 (403M) [application/octet-stream]
Saving to: 'onnetpp-6.4.0-linux-x86_64.tgz'

onnetpp-6.4.0-linux-x86_64.tgz      100%=====>] 403.29M   5.69MB/s   tn 70s

2026-10-07 03:44:05 (5.78 MB/s) - 'onnetpp-6.4.0-linux-x86_64.tgz' saved [422878463|422878463]

student@ubuntu:~/Downloads$ nano manet.tcl
student@ubuntu:~/Downloads$ ls
manet.tcl  onnetpp-6.4.0-linux-x86_64.tgz
student@ubuntu:~/Downloads$ ns manet.tcl
When configured, ns found the right version of tcsh in /usr/bin/tcsh.8.6
but it doesn't seem to be there anymore, so ns will fail back on running the first tcsh in your path. The first one of tcsh may break the test suites. Reconfigure and rebuild ns if this is a problem
n.
num_nodes is set 5
INITIALIZE THE LIST xListHead
Simulation Started...
channel.cc:sendup - Calc highestAntennaZ and distCST_
highestAntennaZ = 1.5, distCST_ = 550.0
ORTING LISTS = TRUE!
Simulation completed.
Trace file: manet.tr
NAM file: manet.nam
student@ubuntu:~/Downloads$
```



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